PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

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| **Programming Fundamentals Lab 05** | **Instructor: Mr. Muhammad Hamza** |
| **Program: AI- Fall 2025** | **Date: 6th November, 2025** |
| **Marks: 10** | **Lab Engineer: Qandil Fatima** |
| **Student Name: Eshal Ayub** | **Reg. No: B25F1061AI100** |

**Instructions:**

- Submit the file with your names following your registration numbers like **AI001_Name.**
- **Copy Paste work will be given no marks.**

# Lab Task

1. C++ program to Print Table of any Number.

   The concept of generating **table of any number** is multiply particular number from 1 to 10.

   num * 1

   num * 2

   num * 3

   num * 4

   num * 5

   num * 6

   num * 7

   num * 8

   num * 9

   num * 10

[*] Untitled1 Untitled2.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cout << "Enter a number to print its table: ";
    cin >> n;

    for(int i = 1; i <= 10; i++)
    {
        cout << n << " x " << i << " = " << n * i << endl;
    }

    return 0;
}
```

Output:

C:\Users\DELL PRECision 7550

```
Enter a number to print its table: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

--------------------------------
Process exited after 5.177 seconds with return value 0
Press any key to continue . . .
```

2. Find power of any number using for loop.

[*] Untitled1 Untitled2.cpp [*] Untitled3

```cpp
#include <iostream>
using namespace std;

int main() {
    int base, exponent;
    long long result = 1;

    cout << "Enter base: ";
    cin >> base;
    cout << "Enter exponent: ";
    cin >> exponent;

    for(int i = 1; i <= exponent; i++) {
        result *= base;
    }

    cout << base << "^" << exponent << " = " << result << endl;
    return 0;
}
```

Output:

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```
Enter base: 2
Enter exponent: \
2^0 = 1

--------------------------------
Process exited after 5.428 seconds with return value 0
Press any key to continue . . .
```

3. Sum of Natural Numbers using loop.

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | [*] Untitled4

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter a positive number: ";
    cin >> n;

    for(int i = 1; i <= n; i++)
    {
        sum += i;
    }

    cout << "Sum of first " << n << " natural numbers is " << sum << endl;
    return 0;
}
```

Output:

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```
Enter a positive number: 4
Sum of first 4 natural numbers is 10

--------------------------------
Process exited after 6.036 seconds with return value 0
Press any key to continue . . .
```

4. C++ program to find sum of digits of a number.

Sum of digits means add all the digits of any number, for example we take any number like 358. Its sum of all digit is 3+5+8=16. Using given code we can easily write cpp program.

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | [*] Untitled4.cpp | [*] Untitled5

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, sum = 0;
    cout << "Enter a number: ";
    cin >> num;

    while(num != 0) {
        sum += num % 10;
        num /= 10;
    }

    cout << "Sum of digits = " << sum << endl;
    return 0;
}
```

Output:

```
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Enter a number: 6
Sum of digits = 6

--------------------------------
Process exited after 2.379 seconds with return value 0
Press any key to continue . . .
```

5. Program to Generate Factorial a Certain Number

Factorial of any number is the product of an integer and all the integers below it for example factorial of 4 is 4! = 4 * 3 * 2 * 1 = 24.

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | [*] Untitled4.cpp | Untitled5.cpp | [*] Untitled6

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    long long factorial = 1;

    cout << "Enter a positive number: ";
    cin >> n;

    for(int i = 1; i <= n; i++) {
        factorial *= i;
    }

    cout << "Factorial of " << n << " = " << factorial << endl;
    return 0;
}
```

Output:

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```
Enter a positive number: 8
Factorial of 8 = 40320

--------------------------------
Process exited after 4.596 seconds with return value 0
Press any key to continue . . .
```

6. Write a C++ program that will print the following pattern.

```
*

**

***

****

*****

******

*******
```

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | [*] Untitled4.cpp | Untitled5.cpp | Untitled6.cpp | [*] Untitled7

```cpp
#include <iostream>
using namespace std;

int main() {
    for(int i = 1; i <= 7; i++) {
        for(int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

Output:

C:\Users\DELL PRECision 7550

```
*
**
***
****
*****
******
*******

--------------------------------
Process exited after 0.08049 seconds with return value 0
Press any key to continue . . .
```

7. Write a C program to input month number and print total number of days in month using switch...case.

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | [*] Untitled4.cpp | Untitled5.cpp | Untitled6.cpp | Untitled7.cpp | [*] Untitled8

```cpp
#include <iostream>
using namespace std;

int main() {
    int month;
    cout << "Enter month number (1-12): ";
    cin >> month;

    switch(month) {
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            cout << "31 days";
            break;
        case 4: case 6: case 9: case 11:
            cout << "30 days";
            break;
        case 2:
            cout << "28 or 29 days (depending on leap year)";
            break;
        default:
            cout << "Invalid month number!";
    }
    return 0;
}
```

Output:

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```
Enter month number (1-12): 7
31 days
--------------------------------
Process exited after 7.701 seconds with return value 0
Press any key to continue . . .
```

8. Write a C program to input an alphabet and check whether it is vowel or consonant using switch case.

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | [*] Untitled4.cpp | Untitled5.cpp | Untitled6.cpp | Untitled7.cp

```cpp
#include <iostream>
using namespace std;

int main() {
    char ch;
    cout << "Enter an alphabet: ";
    cin >> ch;

    switch(ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout << ch << " is a vowel.";
            break;
        default:
            cout << ch << " is a consonant.";
    }
    return 0;

}
```

Output:

```
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Enter an alphabet: b
b is a consonant.
--------------------------------
Process exited after 6.211 seconds with return value 0
Press any key to continue . . .
```